

Lesson 4.8: Color in the Home

Unit: Unit 4, Spaces We Live In (Environmental Design and Management) **Topic:** 4.2 Design: Elements, Principles, Color **Grade:** 6 to 8 (written at grade 7; see Differentiation for the grade 6 support and grade 8 stretch) **Days:** 2 class periods of 40 minutes (days 101 and 102 of the year) **Room:** FACS lab (three stations with sink access) or any classroom with tables that can be covered (a no-food version is below)

This is my Color in the Home lesson from my certification coursework, re-leveled from grades 9 to 12 and one 50-minute period to grades 6 to 8 and two 40-minute periods. The do now, the three-rooms reveal, the frosting mix, the Room Cards, the paper-plate wall panel, the 30-second pitch, and the thumbs vote are kept. The standards table is replaced with the correct middle-level citations, and the rubric is rewritten on the Lab Rubric pattern.

Standards

- NYS module line: NYS EDM 2. Environment and Interior Design c) Incorporate the elements and principles of design into a design plan
- NYS module line: NYS EDM 2. Environment and Interior Design a) Design shared environments and living space zones (for home, school, work, and/or community) which show respect for individual privacy needs and appreciation of large group social needs
- CTE theme line: CTE PSI 2. Design Process (Proactive) a) Implement a formal design process to solve a given problem by b. Defining criteria that must be met through the finished design; c. Defining constraints that must be adhered to; h. Optimizing the solution for best form and function
- CTE theme line: CTE HSW 5. Social-Emotional Learning a) Discuss how personal habits, environment, and heredity affect social-emotional health
- NYS FACS Learning Standard 2 (Intermediate), Key Idea: Students will know the basic principles of home and community safety. They can demonstrate the skills necessary to maintain their homes and workplaces in a safe and comfortable condition. They can provide a safe and nurturing environment for themselves and others.; Performance Indicator: Students know the basics of managing a safe and healthy home.
- CDOS Standard 3a Universal Foundation Skills: Thinking Skills
- National FCS 3.0: 11.2.1 Judge how the elements and principles of design are used in homes and in commercial and residential interiors.
- National FCS 3.0: 11.2.2 Examine how design elements and principles affect how a person feels in a space.
- National FCS 3.0: 11.2.3 Examine how design elements and principles shape how a space looks and how it works.

Note on the theme line: the unit plan called for an HSW 1. Health Practices line on well-being if one fits the color-and-mood idea. None does; HSW 1 defines the dimensions of wellness and health behaviors. HSW 5. a), which names environment as a factor in social-

emotional health, is the closest verbatim line, so it is cited instead. Note on the NYS FACS PI: my original lesson quoted the Commencement-level performance indicator (“apply housing principles (e.g., design and safety) to meet the needs of family members of all ages and abilities”). This is a grade 6 to 8 course, so the Intermediate key idea and PI B are quoted here. Note on the National FCS lines: 11.2.1, 11.2.2, and 11.2.3 are the lines I cited in the original; their 3.0 text is confirmed in 02 Standards/STANDARDS SOURCES - web research.md. The original also cited 11.1.4 and 11.6.5; they are dropped because this lesson does not teach careers or client resources.

Enduring understanding and essential question

Big idea: Color is a tool we use to change how people feel inside a home. The color wheel is the same tool a pastry chef uses to set a table, a fashion designer uses to build an outfit, and an interior designer uses to make a family feel at home. **Essential question:** How does a designer use color to match a room’s mood and its job, and what happens when the family in the room needs something the mood does not give?

Objectives

By the end of this lesson, students will be able to:

1. (Low: remember or understand) Given a blank color wheel, identify the three primary colors and the three secondary colors, and label the warm side and the cool side.
2. (Mid: apply or analyze) Given a Room Card with a room and a mood, apply color theory to choose and mix, from three primaries and white, at least one secondary, one tint, and one shade that fit the assigned mood, and paint them on a wall panel.
3. (High: evaluate or create) In a 30-second pitch, justify the team’s color choices using the family, the mood, and the function of the room, with the words primary, secondary, tint or shade, and warm or cool used correctly.

Vocabulary

Tier 2 (general academic): mix, blend, mood, function, palette, scheme, well-being Tier 3 (FACS): color wheel, primary colors, secondary colors, tertiary colors, tint (color plus white), shade (color plus a touch of a dark color), warm colors, cool colors, interior design, FACS. These are the 17 terms from my own vocabulary cards; see the unit vocabulary list for the two-language table, which needs a native speaker check before printing.

Materials and setup

Three stations of seven students. Each station splits into two sub-teams (four and three) that share the station’s frosting and gel but paint their own wall panel. Six panels in all: four Room Cards plus two Challenge Cards.

Per station:

- One 16-ounce tub of plain white vanilla frosting (the cheapest store brand; check the label for the top allergens and choose a dairy-free and nut-free tub if the roster needs it)
- One set of gel food coloring: red, yellow, blue, plus brown if the set has it (the shade is made with a touch of blue or brown, not black; black gel turns everything gray)
- Paper plates: two per sub-team (one mixing plate, one wall panel) plus one small plate per student for a practice swatch: eleven per station
- Plastic spoons: one per student plus two spares: nine per station
- Parchment paper to cover the table: about three feet
- Wet wipes: one per student, plus a roll of paper towels per station
- One trash bag per station
- Two Room Cards or one Room Card and one Challenge Card per station (from Handout 04.08 - Color Wheel Room Cards and Pitch Sheet.md)

For the whole class:

- Handout, one per student: Handout 04.08 - Color Wheel Room Cards and Pitch Sheet.md (the blank color wheel, the three-things notes, the Room Cards, the pitch sheet, the vocabulary cards)
- Index cards for the Day 1 do now and the Day 2 3-2-1
- The demo kit for Day 1: one tub of frosting, the gel set, three paper plates, three spoons
- Slides: Slides 04.08 - Color in the Home (Day 1 and Day 2 labeled)
- The rubric below, printed on a clipboard with the group scoring grid from Rubric 01 - Lab Rubric.md
- Timer visible to the class

Setup notes: the allergy check is done at the end of Lesson 4.7 (slide 20) and again against the roster before you buy frosting. The rule is “design, not food”: nobody eats, nobody licks a spoon, and any student who cannot touch frosting for any reason uses the tempera version at the same station with no comment. Cover every table with parchment before the Day 2 bell. Put the materials cart in the center of the room so every station can reach it without crossing the room. Budget: at 2026 supermarket prices, one station runs about \$9 to \$11 (frosting about \$2.50, the gel set about \$5, plates about \$0.90, spoons about \$0.35, parchment about \$0.30, wipes about \$0.35); three stations plus the demo tub is about \$32 to \$36 for the class. A set of tempera paint in red, yellow, blue, white, and brown is about \$12 and lasts the year.

Pre-assessment

The day before (the end of Lesson 4.7), the board asks: “Think about your favorite room in your own home. What color is it? Is that color part of why you like the room?” Students answer on an index card on the way out. Then, at the start of Day 1, three versions of the same room are projected in three paint palettes (blues, reds, yellows) for a hand vote: which feels calmest, which feels most exciting? The vote shows where students start before any teaching.

Do now and hook (Time: 5 min)

Day 1. On the board as students enter: “On your index card, list two colors you would put in a CALM room at home, and two colors you would put in an ENERGETIC room. One sentence each on why.”

Hook: reveal that the three rooms on the screen are the same room and the only thing that changed was the paint. Then hold up three squeeze bottles of gel (red, yellow, blue) and a tub of white frosting. “In FACS we use the same color wheel that pastry chefs, fashion designers, and interior designers use. Tomorrow you are going to paint your room with frosting.”

Day 2. On the board (4 minutes today): “In your own words: what is a tint? What is a shade? Then write the first color your team will mix and exactly how you will make it.” Debrief in one minute by cold calling one student per station.

Procedure

Day 1

Step	Teacher will	Students will	Time
1. Direct instruction: color wheel anatomy	Project the color wheel. Name the three primaries: red, yellow, blue. Show how the secondaries come from mixing two primaries: red plus yellow is orange, yellow plus blue is green, red plus blue is purple. Show one tertiary (red plus orange is red-orange) and say the pattern. Draw the line across the wheel: warm side (reds, oranges, yellows) and cool side (blues, greens, purples). Pass out the blank wheel on the handout.	Fill in the color wheel: label the primaries, the secondaries, the warm side, and the cool side. On the index card, write one place in a home besides a wall where color matters: the dinner table, a kid’s room, a holiday meal.	8 min
2. Direct instruction: tint, shade, and	One slide on color and mood: warm	Write three things on the handout, Part	8 min

mood	colors push energy and appetite (kitchens, dining rooms, game rooms); cool colors calm and focus (bathrooms, bedrooms, offices). Tints (color plus white) soften. Shades (color plus a touch of a dark color) add weight. Connect to yesterday's principles: a tint on a big wall keeps visual weight low; a shade on one wall makes a focal point. Then the state's idea in plain words: the environment you live in affects how you feel; a room's color is part of that environment.	2: how to make a secondary color, how to make a tint, how to make a shade. Turn to a partner and predict what white frosting plus blue plus red makes; check the wheel.	
3. Demo: the live frosting mix	At the front on a paper plate: scoop a spoon of white frosting, add one drop of red and one drop of yellow, mix to orange. Add more white to lighten it into a peach tint. Stir in the smallest touch of blue (or brown) to deepen it into a brick shade. Show what happens when the "touch" is a drop: it goes gray. Say the rules for tomorrow: every	Watch, predict each step out loud before it happens, write the result of each step on the handout.	10 min

	station gets one tub of white frosting and red, yellow, and blue gel (plus brown if we have it). Only those colors. Nobody eats. Nobody licks a spoon. Parchment stays on the table. Everything goes in the trash bag, not the sink.		
4. Practice: Room Cards and Part A planning	Hand each sub-team a Room Card or a Challenge Card. Room Cards: Bathroom, calm. Kitchen, warm and welcoming. Game Room, energetic. Home Office, focused. Challenge Cards: Bedroom for a grandparent with low vision, restful but easy to see; Shared bedroom for two kids who fight, calm enough to sleep, fun enough to want to be there. Task for the rest of today: read the card, discuss the family who uses the room, the mood it needs, and which colors on the wheel get there. Write the one-sentence justification draft on the pitch sheet: "We chose ___ for the ___ so the family can	Read the card, plan the palette (one secondary, one tint, one shade at least), draft the sentence.	5 min

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Day 2

Step	Teacher will	Students will	Time
1. Safety and setup	Say the rules once more: design, not food; no tasting, no licking; hands washed before and after; parchment stays; trash bag, not sink; only the colors on the table. Assign the station roles: Station Manager (materials and the timer), Mixer (holds the mixing plate), Painter (the wall panel), Recorder (the pitch sheet), and at seven students a Second Mixer, a Second Painter, and a Cleaner. Check that any student on the tempera version has the paint and a plate.	Wash hands, take a role, open the handout to the pitch sheet.	3 min
2. Lab: mix and paint the wall panel	Start the timer. Circulate to every station at least twice with the rubric clipboard. Push the words: primary, secondary, tint, shade, warm, cool, mood, function. Ask the deepen questions below. At twelve minutes, call "last color." At seventeen, "spoons	Mix the planned colors on the mixing plate with the spoons; make at least one secondary, one tint, and one shade; "paint" the wall panel to show the wall, the floor, and one piece of furniture in the assigned room; each student makes a practice swatch on	17 min

	down.”	the small plate first. The Recorder finishes the justification sentence on the pitch sheet.	
3. Share: the 30-second pitch and the thumbs vote	Each sub-team (six in all) holds up the wall panel and gives a 30-second pitch: room, mood, family, colors used, and why. After each pitch, one deepen question from the teacher. Then the class votes: thumbs up if the colors fit the room and the family, thumbs sideways if not sure. Score Objective 3 on the clipboard as each pitch runs.	Pitch, listen, vote on the other five panels.	8 min
4. Clean up	Run the clean phase like a kitchen lab: scrape frosting into the trash bag, fold the parchment in on itself and bag it, wipe the table, wash hands, station check before anyone sits. Collect the wall panels and pitch sheets for the rubric.	Clean the station, pass the station check.	4 min

Questions to ask

Monitor understanding:

- What are the three primary colors?
- How do you make orange? Green? Purple?
- What is the difference between a tint and a shade?

- Which side of the wheel is warm? Which is cool?
- Look at your mixing plate. Which color did you make first, and how did you get there?

Deepen learning:

- Your room needs to feel calm for the family. Why does adding white to your color help with that?
- How would your colors change if the family had a newborn in this room?
- The grandparent with low vision needs to see the edge of the bed and the door. Does a calm tint help her or hurt her? What is the fix?
- If you were a pastry chef plating dessert for this same room's mood, would your colors change? Why or why not?
- Why is it useful for a FACS professional to know the whole wheel can be mixed from three primaries?

Check for understanding

Day 1: walk the room during Step 1 and check every student's labeled wheel before Step 2; any wheel missing a primary or a secondary gets a quick correction in place. Day 2: the rubric clipboard during Step 2: for each student, a yes or no on "used the words," "made a tint and a shade on the mixing plate," and "the wall panel shows the wall, the floor, and one piece of furniture." The pitch is the Objective 3 check.

Closure (Time: 4 min)

Day 1. Exit card: the first color your team will mix tomorrow, exactly how you will make it, and one word for the mood of your room. Three students read theirs. Any team whose first color is "purple, with red and yellow" gets a wheel back on their desk tomorrow.

Day 2. 3-2-1 exit card from my original lesson: 3 colors you mixed today and what you mixed to get them; 2 ways color affects how a family lives and feels in a room; 1 question you still have about using the color wheel. Collected at the door. Anything on the cards comes back at the start of Lesson 4.9. Sort into "got it" and "needs more."

Differentiation and supports

- ELL: the Spanish vocabulary cards (17 terms, on the handout, to be checked by a native speaker) sit at every station next to the English; the color wheel is the same in every language, so the wheel itself is the visual; the sentence frame on the pitch sheet is the pitch; a bilingual partner where possible.
- IEP and 504 (general): every group produces the same two things (the panel and the sentence), but a student who struggles with writing gives the justification out loud during the pitch instead of on paper; the Mixer role needs no writing at all; a student who cannot handle frosting (allergy, sensory, or any reason) uses tempera on the same plate with no comment; extended time is built in because the pitch sheet can be finished in Lesson 4.9's first minutes; fidgets allowed; the materials cart in the center means nobody crosses the room.

- Grade 6 support: the card asks for one secondary and one tint (no shade required); the pitch is 20 seconds with the frame read off the sheet.
- Grade 8 stretch: the two Challenge Cards go to grade 8 sub-teams where possible; they must defend a less obvious choice (a calm room that still has high contrast at the edges for low vision) and mix at least one tertiary by mixing two primaries in unequal amounts, as in my original lesson's advanced objective.
- UDL checkpoint used: multiple means of representation (the same content arrives by slide, by demo, and by touch), and multiple means of action and expression (the justification is written or spoken; the panel is painted with frosting or tempera).
- No-kitchen or no-food alternative: tempera paint in red, yellow, blue, white, and brown on the same paper plates with the same spoons or with brushes. Everything else is identical: the wheel, the demo (done with tempera), the cards, the panel, the pitch, the vote. Clean-up moves from the trash bag to the sink.

Assessment

Objective	Standards met	Learning task	Assessment (formative or summative; tool)
1. Identify primaries, secondaries, warm and cool on the wheel	EDM 2. c); National FCS 11.2.1	The labeled color wheel, Day 1 Step 1	Formative; each wheel checked in place before Step 2; summative on the Design Vocabulary Check in Lesson 4.10
2. Choose and mix a secondary, a tint, and a shade that fit the mood; paint the panel	EDM 2. c), 2. a); PSI 2. a) b., c., h.; National FCS 11.2.2, 11.2.3; Std 2 PI B	The lab, Day 2 Step 2	Summative; the rubric below, rows 2 and 4, scored on the clipboard during the lab
3. Justify the choice by family, mood, and function in the pitch	EDM 2. a); HSW 5. a); CDOS 3a Thinking Skills; National FCS 11.2.2	The 30-second pitch, Day 2 Step 3; the 3-2-1	Summative; the rubric below, row 3, scored during the pitch; the exit card is read for two ways color affects a family

The rubric: Color in the Home (Lab Rubric pattern, 16 points)

This is Rubric 01 - Lab Rubric.md with the three objective rows rewritten for this lab and the safety row adapted for a design-not-food lab. Use the group scoring grid from that file (three stations of seven) on the clipboard. Category: Labs and projects (40 percent). The Scope and Sequence lists this as the "Color rubric" formal check for Topic 4.2.

Criteria	Not Yet (1)	Approaching	Meets (3)	Exceeds (4)
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		(2)		
Objective 1 (Low). Identified the three primary and three secondary colors on the wheel and labeled the warm and cool sides	Could not name the primaries or the secondaries; the wheel is mostly blank or wrong	Named some primaries or secondaries; the wheel is partly labeled	Named all three primaries and all three secondaries; the wheel is fully labeled with the warm and cool sides	Meets, and also named at least two tertiary colors and said how each was mixed
Objective 2 (Mid). Chose and mixed colors that fit the assigned room and mood, and painted the panel	Colors do not match the mood, or the team mixed nothing new (primaries straight from the bottle onto the panel)	Some colors match the mood; the team mixed one new color but did not show a clear tint or a clear shade	Most colors match the mood; the mixing plate shows at least one visible tint (color plus white) and one visible shade (color plus a touch of blue or brown); the panel shows a wall, a floor, and one piece of furniture	Meets, and the team also used a tertiary color or a mood-specific mix that shows clear intent for the assigned family (for example, high contrast at the edges for low vision)
Objective 3 (High). Justified the color choice by the family, the mood, and the function of the room in the pitch	No justification, or the pitch does not mention the family, the mood, or the function	The pitch names one of the three (family, mood, or function) but is vague	The pitch names all three (family, mood, function) and uses the words primary, secondary, tint or shade, and warm or cool correctly	Meets, and the pitch connects the choice to a specific family member's need or to a real FACS job (a pastry chef, a designer, a nurse choosing a ward color)
Safety and clean-up (every lab, same row all year, adapted for	A rule broken: frosting or a spoon in a mouth, hands not washed,	Hands washed and rules followed, but one reminder needed (a	Washed hands before and after; kept everything on the parchment;	Meets, and reminded a teammate of a rule before I had to, or

design-not-food)	materials taken from another station, frosting in the sink, or a station left dirty after the check. A 1 here means a conversation with me and a redo of the clean phase; it also zeroes the week's participation and safety score (see the Grading Plan)	spoon set down off the parchment, a plate on the floor, the trash bag not used)	used only the station's colors; scraped, bagged, wiped; passed the station check the first time	cleaned a spill at another station without being asked
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Total: ____ out of 16

How to use: score Objective 1 from the Day 1 wheel; score Objective 2 and Safety on the clipboard during Day 2 Step 2 and Step 4; score Objective 3 during the pitch. Scores are per student, not per sub-team, because roles differ; one student's 1 on safety does not lower a teammate's score. Students copy their four numbers into the FACS folder. Absent students do the written alternative (label a wheel, plan a palette for a Room Card in writing, and write the justification) and are scored on Objectives 1 to 3 out of 12, scaled to 16.

Homework

None. Optional: pick a room at home and write which side of the wheel it is on and whether that fits its job. No photos.

Connections

Art: the art teacher's color wheel is the same wheel; a ten-minute swap where FACS borrows the art room's paint tests and art borrows the "which room" question keeps the two from teaching it twice. Culinary: the frosting is not a joke; pastry chefs use the same gels and the same wheel, and this is the link to Unit 6's decorating lab. Community: a paint store's free color chips are the swatches for Lesson 4.9's mood boards; ask the store for the discontinued chip decks. Health: the mood and environment idea (HSW 5. a)) is a two-minute overlap with the health teacher's stress unit; tell the health teacher which week this runs.

Sources

- NYS Middle Level CTE FACS Content Module 04, Environmental Design and Management (nycotecenter.org, June 2018), lines 2. a) and 2. c). Copy in 02 Standards/NYS Module PDFs/.

- NYS Middle Level CTE Theme Module, Problem Solving and Innovation, line 2. a) with sub-lines b., c., and h.; Theme Module, Health, Safety, and Wellness, line 5. a). Text per 02 Standards/NYS ML CTE FACS Modules - verbatim extract of all twelve.md.
- NYS Learning Standards for Health, Physical Education, and Family and Consumer Sciences (1996), Standard 2, Intermediate. Wording per the crosswalk, Part A.6.
- National Standards for Family and Consumer Sciences Education 3.0 (LEAD FCS Education, 2018), competencies 11.2.1, 11.2.2, and 11.2.3, per 02 Standards/STANDARDS SOURCES - web research.md.
- My own materials: my Color in the Home lesson plan, its rubric, and the vocabulary handouts that went with it. My “Color Me Impressed” and “Mood Maker” packets were read but not reused here.

Teacher notes

- If Day 1 runs short, cut the tertiary example in Step 1 and the newborn question in Step 2. If Day 2 runs short, cut the pitches to 20 seconds and the vote to one thumbs per panel. Never cut the clean phase; a station left with frosting on it is the story the custodian tells the principal.
- Common mistake: too much gel. One drop of gel colors a whole spoon of frosting. Say “a toothpick of gel” and put toothpicks on the cart.
- Common mistake: making the shade with a drop of blue instead of a touch. The demo’s gray moment on Day 1 is on purpose; refer back to it.
- Brown gel is the better shading color for warm rooms (kitchen, game room); blue is better for cool rooms (bathroom, office). If the set has only three colors, blue shades everything and that is fine.
- The low-vision Challenge Card is the one that produces the best pitch every time: the team has to argue against “calm equals pale” and put a dark shade at the door frame and the bed edge. Give it to the team that argues the most.
- Photograph the six panels before they go in the trash. They are the review slides for the Design Vocabulary Check and the color examples for the Design a Room project.